ML Sample Question-Answer

<u>Elaborate yourself if necessary</u>

Q.1.What is Machine learning and how is it different from Deep learning?

Answer:

Machine learning develop programs that can access data and learn from it. Deep learning is the sub domain of the machine learning. Deep learning supports automatic extraction of features from the raw data.

Q.2.What are the different type of machine learning algorithms?

Answer:

- Supervised algorithms: These are the algorithms which learn from the labelled data, e.g. images labelled with dog face or not. Algorithm depends on supervised or labelled data. E.g. regression, object detection, segmentation.
- Unsupervised algorithms: These are the algorithms which learn from the non labelled data, e.g. bunch of images given to make a similar set of images. e.g. clustering, dimensionality reduction etc.
- Semi-Supervised algorithms: Algorithms that uses both supervised and unsupervised data. Majority portion of data use for these algorithms are not supervised data. E.g. anamoly detection.

Q.3.Why we use machine learning?

Answer:

Machine learning is used to make decisions based on data. By modelling the algorithms on the bases of historical data, Algorithms find the patterns and relationships that are difficult for humans to detect. These patterns are now further use for the future references to predict solution of unseen problems.

Q.4.What is the difference between Artificial Intelligence and Machine learning?

Answer:

S.N.	Artificial intelligence	Machine learning
1.	The terminology "Artificial Intelligence" was originally used by **John McCarthy in 1956**, who also hosted the first AI conference.	The terminology "Machine Learning" was first used in **1952 by IBM computer scientist Arthur Samuel**, a pioneer in artificial intelligence and computer games.
2.	AI stands for Artificial intelligence, where intelligence is defined as the ability to acquire and apply knowledge.	ML stands for Machine Learning which is defined as the acquisition of knowledge or skill
3.	AI is the broader family consisting of ML and DL as its components.	Machine Learning is the subset of Artificial Intelligence.
4.	The aim is to increase the chance of success and not accuracy.	The aim is to increase accuracy, but it does not care about; the success
5.	AI is aiming to develop an intelligent system capable of performing a variety of complex jobs. decision-making	Machine learning is attempting to construct machines that can only accomplish the jobs for which they have been trained.

S.N.	Artificial intelligence	Machine learning
6.	It works as a computer program that does smart work.	Here, the tasks systems machine takes data and learns from data.
7.	The goal is to simulate natural intelligence to solve complex problems.	The goal is to learn from data on certain tasks to maximize the performance on that task.
8.	AI has a very broad variety of applications.	The scope of machine learning is constrained.
9.	AI is decision-making.	ML allows systems to learn new things from data.
10.	It is developing a system that mimics humans to solve problems.	It involves creating self-learning algorithms.
11.	AI is a broader family consisting of ML and DL as its components.	ML is a subset of AI.
12.	Three broad categories of AI are : 1. Artificial Narrow Intelligence (ANI) 2. Artificial General Intelligence (AGI) 3. Artificial Super Intelligence (ASI)	Three broad categories of ML are : 1. Supervised Learning 2. Unsupervised Learning 3. Reinforcement Learning

S.N.	Artificial intelligence	Machine learning
13.	AI can work with structured, semi-structured, and unstructured data.	ML can work with only structured and semi-structured data.
14.	AI's key uses include- • Siri, customer service via chatbots • Expert Systems • Machine Translation like Google Translate • Intelligent humanoid robots such as Sophia, and so on.	The most common uses of machine learning- • Facebook's automatic friend suggestions • Google's search algorithms • Banking fraud analysis • Stock price forecast • Online recommender systems, and so on.
15.	AI refers to the broad field of creating machines that can simulate human intelligence and perform tasks such as understanding natural language, recognizing images and sounds, making decisions, and solving complex problems.	ML is a subset of AI that involves training algorithms on data to make predictions, decisions, and recommendations.
16.	AI is a broad concept that includes various methods for creating intelligent machines, including rule-	ML focuses on teaching machines how to learn from data without being explicitly programmed, using

S.N.	Artificial intelligence	Machine learning
	based systems, expert systems, and machine learning algorithms. AI systems can be programmed to follow specific rules, make logical inferences, or learn from data using ML.	algorithms such as neural networks, decision trees, and clustering.
17.	AI systems can be built using both **structured** and **unstructured** dat a, including text, images, video, and audio. AI algorithms can work with data in a variety of formats, and they can analyze and process data to extract meaningful insights.	In contrast, ML algorithms require large amounts of structured data to learn and improve their performance. The quality and quantity of the data used to train ML algorithms are critical factors in determining the accuracy and effectiveness of the system.
18.	AI is a broader concept that encompasses many different applications, including **robotics**, **natural language processing**, **speech recognition**, and **autonomous vehicles**. AI systems can be used to solve complex problems in various fields, such as healthcare, finance, and transportation.	ML, on the other hand, is primarily used for **pattern recognition**, **predictive modeling**, and decision-making in fields such as marketing, fraud detection, and credit scoring.

S.N.	Artificial intelligence	Machine learning
19.	AI systems can be designed to work autonomously or with minimal human intervention, depending on the complexity of the task. AI systems can make decisions and take actions based on the data and rules provided to them.	In contrast, ML algorithms require human involvement to set up, train, and optimize the system. ML algorithms require the expertise of data scientists, engineers, and other professionals to design and implement the system.

Q.5. Define Machine learning and its different types.

Machine learning is a subfield of artificial intelligence (AI) that uses algorithms trained on data sets to create self-learning models that are capable of predicting outcomes and classifying information without human intervention. Machine learning is used today for a wide range of commercial purposes, including suggesting products to consumers based on their past purchases, predicting stock market fluctuations, and translating text from one language to another.

Some of the most common examples of machine learning that we interact include:

- Recommendation engines that suggest products, songs, or television shows to you, such as those found on Amazon, Spotify or Netflix.
- Speech recognition software that allows you to convert voice memos into text.
- A bank's fraud detection services automatically flag suspicious transactions.
- Self-driving cars and driver assistance features, such as blind-spot detection and automatic stopping, improve overall vehicle safety.

Types of Machine learning

1. **Supervised Machine learning**: In supervised machine learning, algorithms are trained on labeled data sets that include tags describing each piece of data. In other words, the algorithms are fed data that includes an "answer key" describing how the data should be interpreted. For example, an algorithm may be fed images of flowers that include tags for each flower type so that it will be able to identify the flower better again when fed a new photograph. Supervised machine learning is often used to create machine learning models used for prediction and classification purposes.
2. **Unsupervised Machine learning:** Unsupervised machine learning uses unlabeled data sets to train algorithms. In this process, the algorithm is fed data that doesn't include tags, which requires it to uncover patterns on its own without any outside guidance. For instance, an algorithm may be fed a large amount of unlabeled user data culled from a social media site in order to identify behavioral trends on the platform.

Unsupervised machine learning is often used by researchers and data scientists to identify patterns within large, unlabeled data sets quickly and efficiently.

3. Semi-supervised machine learning: Semi-supervised machine learning uses both unlabeled and labeled data sets to train algorithms. Generally, during semi-supervised machine learning, algorithms are first fed a small amount of labeled data to help direct their development and then fed much larger quantities of unlabeled data to complete the model. For example, an algorithm may be fed a smaller quantity of labeled speech data and then trained on a much larger set of unlabeled speech data in order to create a machine learning model capable of speech recognition.

Semi-supervised machine learning is often employed to train algorithms for classification and prediction purposes in the event that large volumes of labeled data is unavailable.

4. Reinforcement learning: Reinforcement learning uses trial and error to train algorithms and create models. During the training process, algorithms operate in specific environments and then are provided with feedback following each outcome. Much like how a child learns, the algorithm slowly begins to acquire an understanding of its environment and begins to optimize actions to achieve particular outcomes. For instance, an algorithm may be optimized by playing successive games of chess, which allows it to learn from its past successes and failures playing each game.

Reinforcement learning is often used to create algorithms that must effectively make sequences of decisions or actions to achieve their aims, such as playing a game or summarizing an entire text.

Deep learning vs. Machine learning

Machine learning and deep learning are both types of AI. In short, machine learning is AI that can automatically adapt with minimal human interference. Deep learning is a subset of machine learning that uses artificial neural networks to mimic the learning process of the human brain.

S.N	Machine learning	Deep learning
1	A subset of AI.	A subset of machine learning.
2	Can train on smaller data sets.	Requires large amounts of data.
3	Requires more human intervention to correct and learn.	Learns on its own from environment and past mistakes.
4	Shorter training and lower accuracy.	Longer training and higher accuracy.
5	Makes simple, linear correlations.	Makes non-linear, complex correlations.

S.N	Machine learning	Deep learning
6	Can train on a CPU (central processing unit).	Needs a specialized GPU (graphics processing unit) to train.

Q.6. Explain the difference between supervised and unsupervised machine learning?

***Answer*:** In supervised machine learning algorithms, we have to provide labeled data, for example, prediction of stock market prices, whereas in unsupervised we do not have labeled data where we group the unlabeled data, for example, conducting market segmentation.

Q.7. Explain the difference between KNN and K-Means clustering?

Answer: K-Nearest Neighbors is a supervised machine learning algorithm where we need to provide the labeled data to the model it then classifies the points based on the distance of the point from the nearest points. Whereas, on the other hand, K-Means clustering is an unsupervised machine learning algorithm thus we need to provide the model with un-labelled data and this algorithm classifies points into clusters based on the mean of the distances between different points.

Q.8.What is the difference between classification and regression?

Answer: Classification is used to produce discrete results, classification is used to classify data into some specific categories .for example classifying e-mails into spam and non-spam categories. Whereas, we use regression analysis when we are dealing with continuous data, for example predicting stock prices at a certain point of time.

Q.9. How to ensure that your model is not overfitting?

Answer: Keep the design of the model simple. Try to reduce the noise in the model by considering fewer variables and parameters. Cross-validation techniques such as

K-folds cross-validation help us keep overfitting under control. Regularization techniques such as LASSO help in avoiding overfitting by penalizing certain parameters if they are likely to cause overfitting.

Q.10.What are the different sets in which we divide any dataset for Machine Learning?

Answer: For any ML application, we divide our dataset into three segments namely 'Training Set', 'Validation Set' & 'Testing Set'. Training Set is used for training the ML model, Validation Set is used for Hyper-parameter tuning and Testing Set is used for testing the model to see how well it is performing.

Q.11. List the main advantage of Naive Bayes?

Answer: A Naive Bayes classifier converges very quickly as compared to other models like logistic regression. As a result, we need less training data in the case of a naive Bayes classifier.

Q.12. Explain Ensemble learning.

Answer: In ensemble learning, many base models like classifiers and regressions are generated and combined together so that they give better results. It is used when we build component classifiers that are accurate and independent. There are sequential as well as parallel ensemble methods.

Q.13. Explain Dimensionality Reduction in machine learning.

Answer: Dimensionality Reduction is the method of reducing the number of dimensions of any dataset by reducing the number of features. It is important because as we move into higher dimensions, the data points start becoming equidistant from each other which can affect the performance of unsupervised ML algorithms which use Euclidean distance as the similarity function to classify data points. This is known as the Curse of Dimensionality. Also, it is difficult to visualize data beyond 4 dimensions.

Q.14.What should you do when your model is suffering from low bias and high variance?

Answer: When the model is suffering from low bias and high variance, it is essentially overfitting, where the accuracy of train dataset is much higher than the accuracy of the test dataset. In such a situation, techniques such as Regularization can be used or the model can be simplified by reducing the number of features in the dataset.

Q.15. Explain the differences between random forest and gradient boosting algorithms.

Answer: Random forest uses bagging techniques whereas GBM uses boosting techniques. Random forests mainly try to reduce variance and GBM reduces both bias and variance of a model.

Q.16.What are some real-life applications of clustering algorithms?

Answer: The clustering technique can be used in multiple domains of data science like image classification, customer segmentation, and recommendation engine. One of the most common use is in market research and customer segmentation which is then utilized to target a particular market group to expand the businesses and profitable outcomes.

Q.17. How to choose an optimal number of clusters?

Answer: By using the Elbow method we decide an optimal number of clusters that our clustering algorithm must try to form. The main principle behind this method is that if we will increase the number of clusters the error value will decrease. But after an optimal number of features, the decrease in the error value is insignificant so, at the point after which this starts to happen, we choose that point as the optimal number of clusters that the algorithm will try to form.

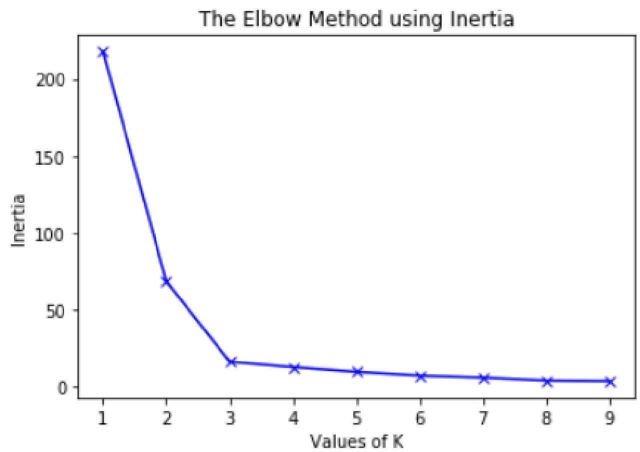

Fig: Elbow Method

Q.18.What is feature engineering? How does it affect the model's performance?

Answer: Feature engineering refers to developing some new features by using existing features. Sometimes there is a very subtle mathematical relation between some features which if explored properly then the new features can be developed using those mathematical operations.

Also, there are times when multiple pieces of information are clubbed and provided as a single data column. At those times developing new features and using them help us to gain deeper insights into the data as well as if the features derived are significant enough helps to improve the models performance a lot**.**

Q.19.What is a Hypothesis in Machine Learning?

Answer: A hypothesis is a term that is generally used in the supervised machine learning domain. As we have independent features and target variables and we try to find an approximate function mapping from the feature space to the target variable that approximation of mapping is known as a hypothesis.

Q.20.What is Overfitting in Machine Learning and how can it be avoided?

Answer: Overfitting happens when the model learns patterns as well as the noises present in the data this leads to high performance on the training data but very low performance for data that the model has not seen earlier.

To avoid overfitting there are multiple methods that we can use:

- Early stopping of the model's training in case of validation training stops increasing but the training keeps going on.
- Using regularization methods like L1 or L2 regularization which is used to penalize the model's weights to avoid overfitting.

Q.21.Why we cannot use linear regression for a classification task?

Answer: The main reason why we cannot use linear regression for a classification task is that the output of linear regression is continuous and unbounded, while classification requires discrete and bounded output values.

If we use linear regression for the classification task the error function graph will not be convex. A convex graph has only one minimum which is also known as the global minima but in the case of the non-convex graph, there are chances of our model getting stuck at some local minima which may not be the global minima. To avoid this situation of getting stuck at the local minima we do not use the linear regression algorithm for a classification task.

Q.22.What is the difference between precision and recall?

Answer: Precision is simply the ratio between the true positives (TP) and all the positive examples (TP+FP) predicted by the model. In other words, precision measures how many of the predicted positive examples are actually true positives. It is a measure of the models ability to avoid false positives and make accurate positive predictions.

$$Precision = TP / TP + FP$$

But in the case of a recall, we calculate the ratio of true positives (TP) and the total number of examples (TP+FN) that actually fall in the positive class. Recall measures how many of the actual positive examples are correctly identified by the model. It is a measure of the model's ability to avoid false negatives and identify all positive examples correctly.

Recall=TP / TP + FN

Q.23.What is data leakage and how can we identify it?

Answer: If there is a high correlation between the target variable and the input features then this situation is referred to as data leakage. This is because when we train our model with that highly correlated feature then the model gets most of the target variable's information in the training process only and it has to do very little to achieve high accuracy. In this situation, the model gives pretty decent performance both on the training as well as the validation data but as we use that model to make actual predictions then the model's performance is not up to the mark. This is how we can identify data leakage.

Q.24.What are some of the hyper-parameters of the random forest regression which help to avoid overfitting?

Answer: The most important hyper-parameters of a Random Forest are:

max_depth – Sometimes the larger depth of the tree can create overfitting. To overcome it, the depth should be limited.

n-estimator – It is the number of decision trees we want in our forest.

min_sample_split – It is the minimum number of samples an internal node must hold in order to split into further nodes.

max_leaf_nodes – It helps the model to control the splitting of the nodes and in turn, the depth of the model is also restricted.

Q.25.What is the bias-variance tradeoff?

Answer: Bias refers to the difference between the actual values and the predicted values by the model. Low bias means the model has learned the pattern in the data and high bias means the model is unable to learn the patterns present in the data i.e. the under fitting.

Variance refers to the change in accuracy of the model's prediction on which the model has not been trained. Low variance is a good case but high variance means that the performance of the training data and the validation data vary a lot.
If the bias is too low but the variance is too high then that case is known as overfitting. So, finding a balance between these two situations is known as the bias-variance trade-off.

Q.26.How can you conclude about the model's performance using the confusion matrix?

Answer: Confusion matrix summarizes the performance of a classification model. In a confusion matrix, we get four types of output (in case of a binary classification problem) which are TP, TN, FP, and FN. As we know that there are two diagonals possible in a square and one of these two diagonals represents the numbers for which our model's prediction and the true labels are the same. Our target is also to maximize the values along these diagonals. From the confusion matrix, we can calculate various evaluation metrics like accuracy, precision, recall, F1 score, etc.

Q.27.What is the difference between stochastic gradient descent (SGD) and gradient descent (GD)?

Answer: In the gradient descent algorithm train our model on the whole dataset at once. But in Stochastic Gradient Descent, the model is trained by using a mini-batch of training data at once. If we are using SGD then one cannot expect the training error to go down smoothly. The training error oscillates but after some training steps, we can say that the training error has gone down. Also, the minima achieved by using GD may vary from that achieved using the SGD. It is observed that the minima achieved by using SGD are close to GD but not the same.

Q.28. Explain the working principle of SVM.

Answer: A data set that is not separable in different classes in one plane may be separable in another plane. This is exactly the idea behind the SVM in this a low dimensional data is mapped to high dimensional data so, that it becomes separable in the different classes. A hyperplane is determined after mapping the data into a higher dimension which can separate the data into categories. SVM model can even learn non-linear boundaries with the objective that there should be as much margin as possible between the categories in which the data has been categorized. To perform this mapping different types of kernels are used like radial basis kernel, Gaussian kernel, polynomial kernel and many others.

Q.29.What is the difference between the k-means and k-means++ algorithms?

Answer: The only difference between the two is in the way centroids are initialized. In the k-means algorithm, the centroids are initialized randomly from the given points. There is a drawback in this method that sometimes this random initialization leads to non-optimized clusters due to maybe initialization of two clusters close to each other.

To overcome this problem k-means++ algorithm was formed. In k-means++, the first centroid is selected randomly from the data points. The selection of subsequent centroids is based on their separation from the initial centroids. The probability of a point being selected as the next centroid is proportional to the squared distance between the point and the closest centroid that has already been selected. This guarantees that the centroids are evenly spread apart and lowers the possibility of convergence to less-than-ideal clusters. This helps the algorithm reach the global minima instead of getting stuck at some local minima.

Q.30. Explain the working procedure of the XGB model.

Answer: XGB model is an example of the ensemble technique of machine learning in this method weights are optimized in a sequential manner by passing them to the decision trees. After each pass, the weights become better and better as each tree tries to optimize the weights, and finally, we obtain the best weights for the problem at hand. Techniques like regularized gradient and mini-batch gradient descent have been used to implement this algorithm so, that it works in a very fast and optimized manner.

Q.31.What is the difference between k-means and the KNN algorithm?

Answer: k-means algorithm is one of the popular unsupervised machine learning algorithms which is used for clustering purposes. But the KNN is a model which is generally used for the classification task and is a supervised machine learning algorithm. The k-means algorithm helps us to label the data by forming clusters within the dataset.

Simple Questions

1. What is the difference between supervised and unsupervised learning?
2. What is overfitting and how can it be avoided?
3. Explain the bias-variance tradeoff.
4. What is a confusion matrix?

Intermediate Questions

1. What is feature engineering and why is it important?
2. How do you handle missing data in a dataset?
3. What is cross-validation and why is it used?
4. Explain the concept of regularization and its types.
5. What are Different Kernels in SVM?
6. Explain the Difference Between Classification and Regression? What is Cross-Validation?
7. What are Support Vectors in SVM? Explain SVM Algorithm in Detail
8. What is PCA? When do you use it?
9. What is ‘Naive’ in a Naive Bayes?

10. What is Unsupervised Learning?

11. What is Supervised Learning?

12. What are Different Types of Machine Learning algorithms?

Advanced Questions

1. What is a support vector machine (SVM) and how does it work?

2. Explain the difference between bagging and boosting.

3. What is a convolutional neural network (CNN)?

4. How does gradient descent work?

Scenario-Based Questions

1. How would you approach a machine learning problem where the dataset is highly imbalanced?

2. Describe a situation where you had to choose between different machine learning models. How did you make your decision?

Advanced Machine Learning Questions

1. What is F1 score? How would you use it?

2. Define Precision and Recall?

3. How to Tackle Overfitting and Underfitting?

4. What is a Neural Network?

5. What are Loss Function and Cost Functions? Explain the key Difference Between them.

6. What is Ensemble learning?

7. How do you make sure which Machine Learning Algorithm to use?

8. How to Handle Outlier Values?

9. What is a Random Forest? How does it work?

10. What is Collaborative Filtering? And Content-Based Filtering?
11. What is Clustering? How can you select K for K-means Clustering?
13. What are Recommender Systems?
14. How do check the Normality of a dataset?
15. Can logistic regression use for more than 2 classes?
16. Explain Correlation and Covariance?
17. What is P-value?
18. What are Parametric and Non-Parametric Models?
19. What is Reinforcement Learning?
20. Difference between Sigmoid and Softmax functions?